A

PROJECT REPORT

ON

“Student Performance Prediction System “

SUBMITTED TO

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

IN PARTIAL FULFILLMENT OF

T.Y.B.Sc.(Computer Science) - Semester VI

SUBMITTED BY
JAY SHANKAR PODDAR
SAURABH SANJAY PATIL
RAHUL KUMAR RAVI RANJAN PRASOON

UNDER THE GUIDANCE OF

PROF.ALHAT SHITAL

PROF.LONDHE MADHURI

DEPARTMENT OF COMPUTER SCIENCE

RAJMATA JIJAU SHIKSHAN PRASARAK MANDAL’S

ARTS, COMMERCE & SCIENCE COLLEGE

LANDEWADI, BHOSARI, PUNE-411039

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Rajmata Jijau Shikshan Prasarak Mandal's

Arts, Commerce & Science College,

Landewadi, Bhosari, Pune – 411039.

(Affiliated to Savitribai Phule Pune University & Approved by Government of Maharashtra)

Department of Computer Science

CERTIFICATE

This is to certify that, _____, a student of ***T.Y.B.Sc.(Computer Science) Sem -VI*** *has* satisfactorily completed the Project _____ , as per the syllabus laid down by the ***Savitribai Phule Pune University*** during the academic year ***2025-2026*** .

Exam Seat No: -

_____	_____	_____
Project Guide	**Head of Dept**	**Principal**

_____	_____
Internal Examiner	**External Examiner**

Date:-

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INTRODUCTION

The Student Performance Prediction and Analysis System is a machine learning-based web application developed to predict a student's academic performance using various academic, behavioral, and lifestyle factors.

Student performance depends on multiple factors such as study hours, sleep patterns, productivity, screen time, and mental condition. However, students often fail to understand how these factors influence their results.

This system analyzes such data and predicts the expected exam score. In addition to prediction, it also provides feedback, improvement suggestions, and performance insights.

By using machine learning techniques, the system helps students make better decisions and improve their academic performance.

Problem Statement

In today's academic environment, students often struggle to identify the factors that affect their performance.

Manual analysis of habits such as study time, sleep, and distractions is difficult and not always accurate. Students usually recognize their mistakes only after poor results are obtained.

There is no simple system available that can analyze student behavior and predict performance in advance.

Therefore, there is a need for an automated system that can analyze student data and provide accurate predictions along with useful insights.

Objective and Goals

Goals:

• To build a system that predicts student exam performance
• To provide practical insights for improving academic habits
• To simplify performance analysis

Objectives:

• To collect student data such as study hours, sleep, and productivity
• To apply machine learning for score prediction
• To provide grade and GPA estimation
• To generate personalized feedback and suggestions
• To improve prediction reliability using validation and adjustment logic

Scope of the System

The system provides a platform where students can input their data and receive predictions along with feedback.

Scope includes:
• Data input through web interface
• Machine learning-based prediction
• Feedback and improvement suggestions
• Validation of input data
• Short-term prediction analysis

The system is intended as a decision-support tool and not a replacement for academic evaluation.

Existing System and Need of the System

Existing System

In the existing system, student performance analysis is mostly done manually by teachers or through self-evaluation.

Students rely on exam results or teacher feedback to understand their performance. This process is time-consuming and may not always provide accurate insights.

There is no system that directly connects student habits with predicted performance.

Need of the System

There is a need for a system that can automatically analyze student data and predict performance.

Such a system can:
• Provide quick and accurate predictions
• Help students understand their weaknesses
• Improve study planning and habits
• Reduce manual effort

This system makes academic analysis easier and more effective.

PROPOSED SYSTEM

he proposed system is a web-based application that allows students to enter their academic and lifestyle data and receive performance predictions.

The system uses machine learning algorithms to analyze the input data and generate results.

Main Functions:

1. **Data Input** – Students enter details such as study hours, sleep, and productivity
2. **Prediction** – System predicts exam score
3. **Feedback** – Provides suggestions for improvement
4. **Result Display** – Shows performance level and insights
5. **Data Storage** – Stores user data for future use

Advantages and Disadvantages

Advantages

• Easy and user-friendly interface
• Provides quick predictions
• Helps students improve performance
• Saves time and effort
• Provides useful insights

Disadvantages

• Accuracy depends on input data
• Cannot replace actual exams
• Limited by dataset quality
• Prediction is not 100% accurate

System Requirements

Software Requirements

• OS: Windows / Linux
• Frontend: HTML, CSS
• Backend: Python (Flask)
• ML Libraries: Scikit-learn, Pandas, NumPy
• Browser: Chrome / Firefox

Hardware Requirements

• Processor: Intel i3 or above
• RAM: Minimum 4 GB
• Storage: Minimum 10 GB

Feasibility Study

Technical Feasibility

The system is developed using Python, Flask, HTML, and CSS, which are widely used and easy to implement. These technologies are suitable for building a web-based application.

Economic Feasibility

The system uses open-source technologies, so the cost of development is low. It does not require expensive hardware or software.

Operational Feasibility

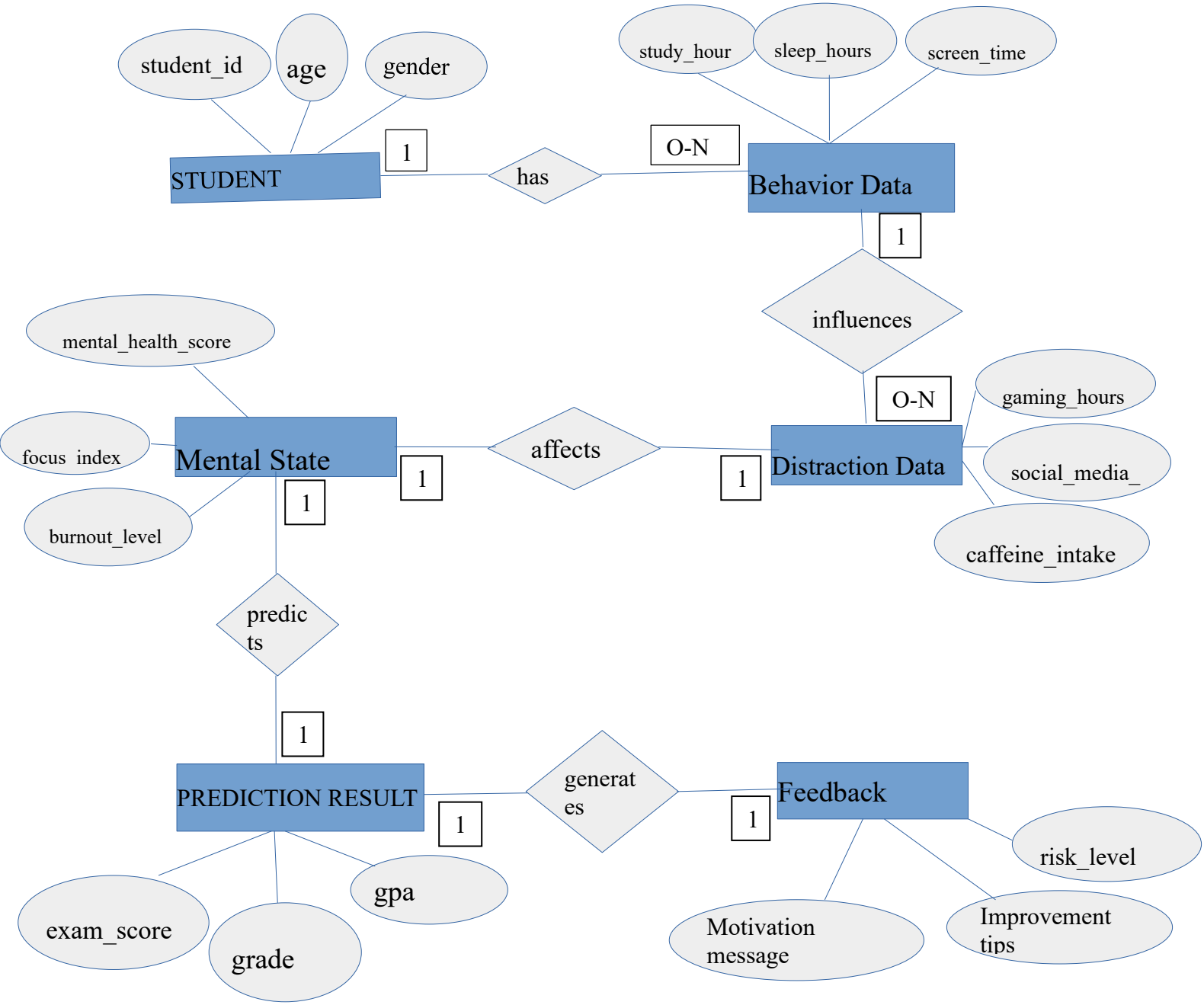
The system is simple to use. Students can easily enter data and get results without technical knowledge, making it user-friendly and practical.

DATA DICTIONARY (Table)

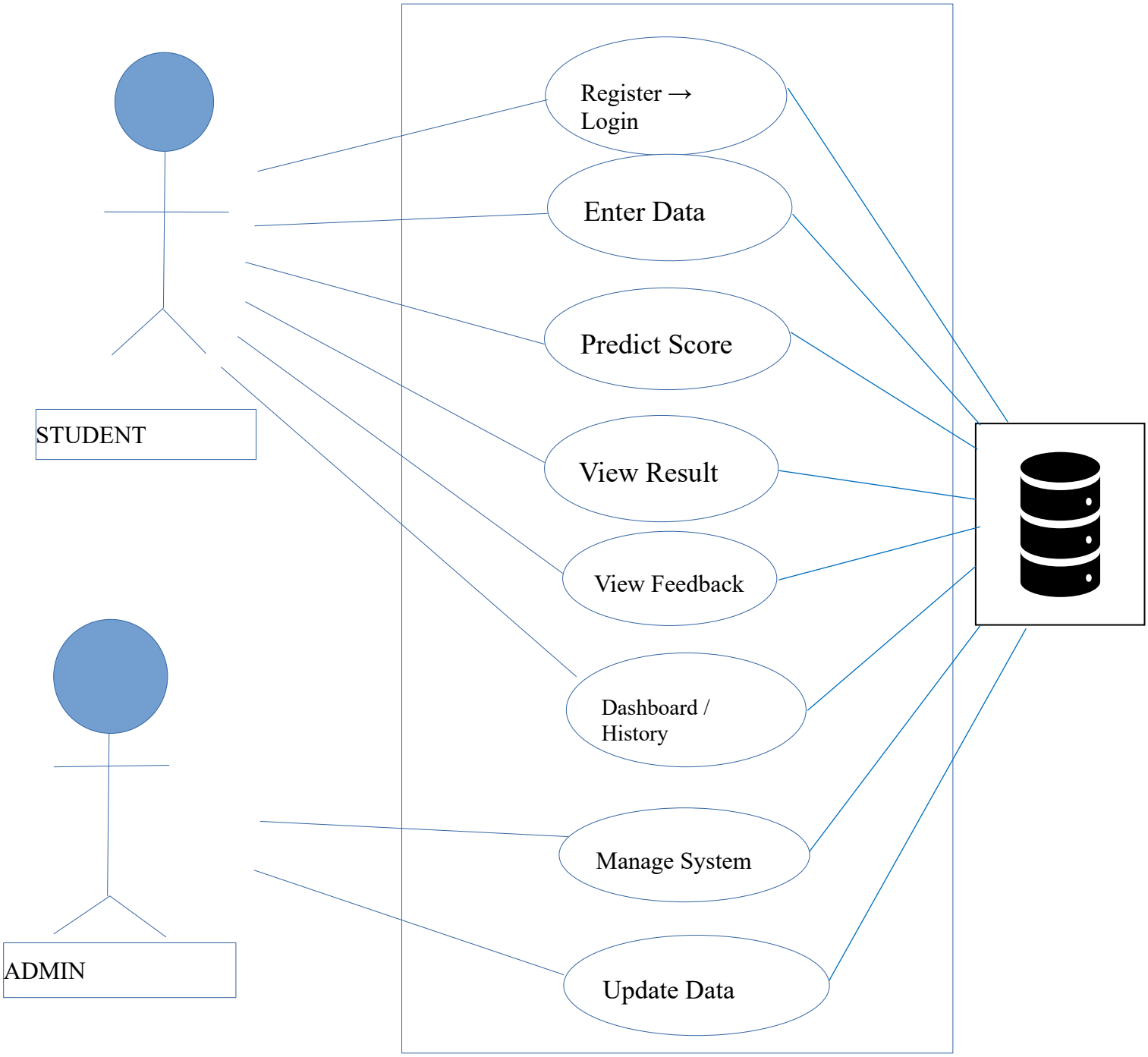
Column	Data Type	Description
id	Integer	Student ID
name	Text	Student Name
age	Integer	Age
study_hours	Float	Study Hours
sleep_hours	Float	Sleep Duration
screen_time	Float	Screen Time
productivity	Float	Productivity Score
burnout	Float	Burnout Level
exam_score	Float	Predicted Score

SYSTEM MODELING

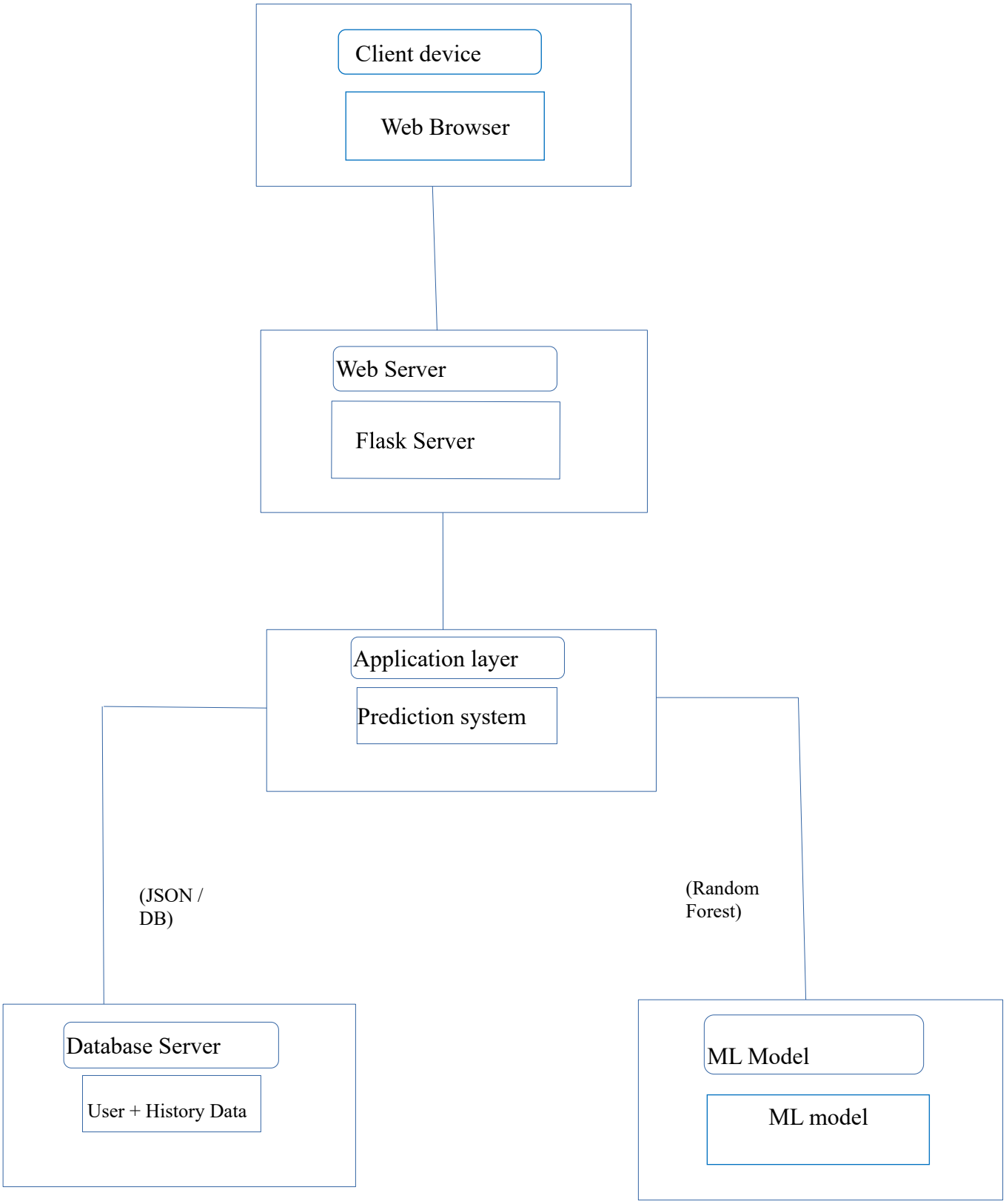
ER DIAGRAM:-



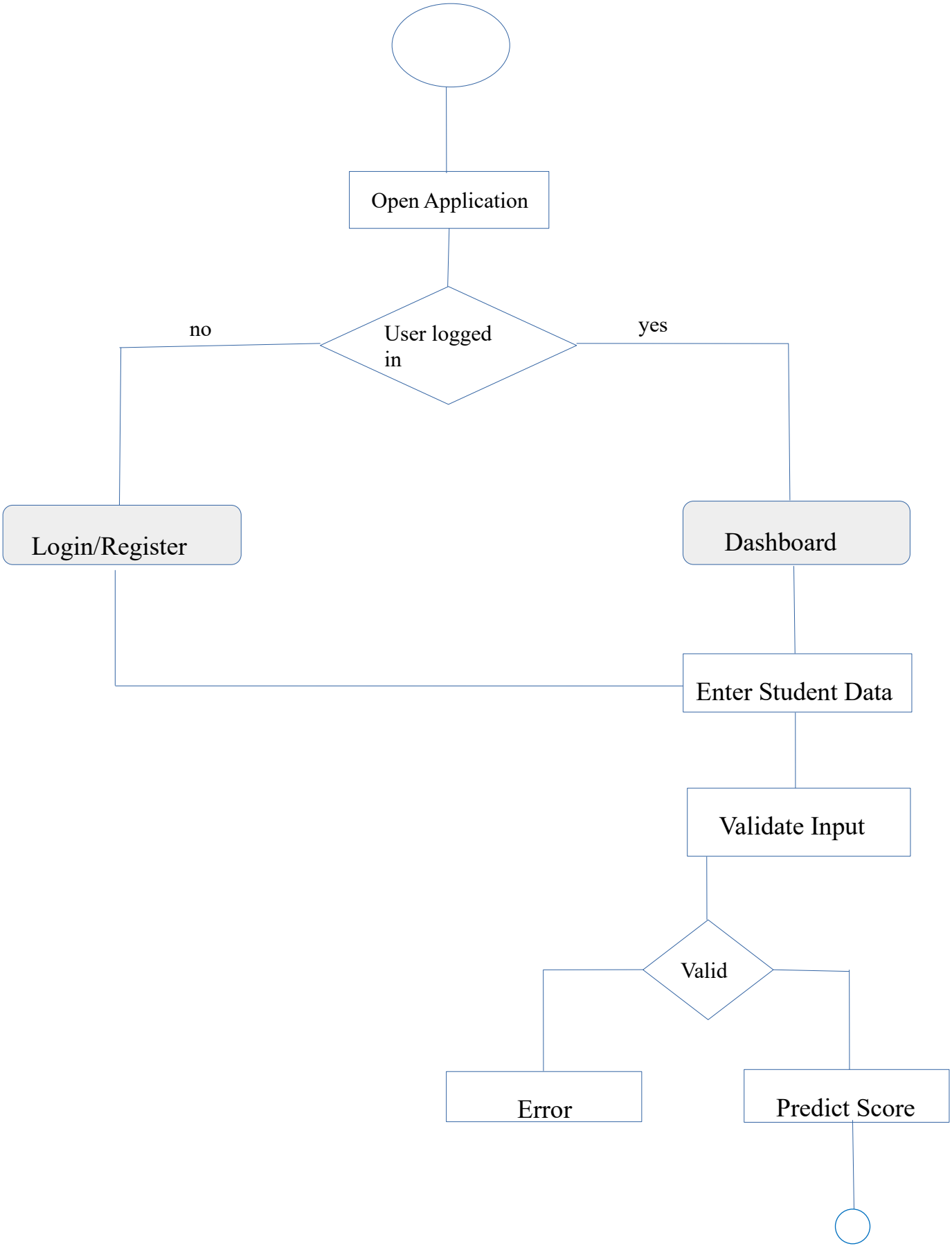
USECASE DIAGRAM:-

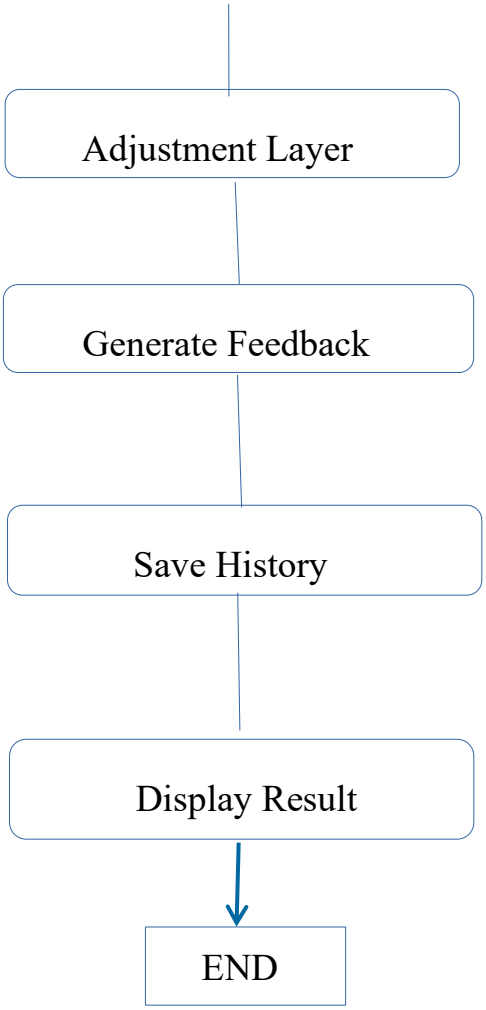


DEPLOYMENT DIAGRAM:-



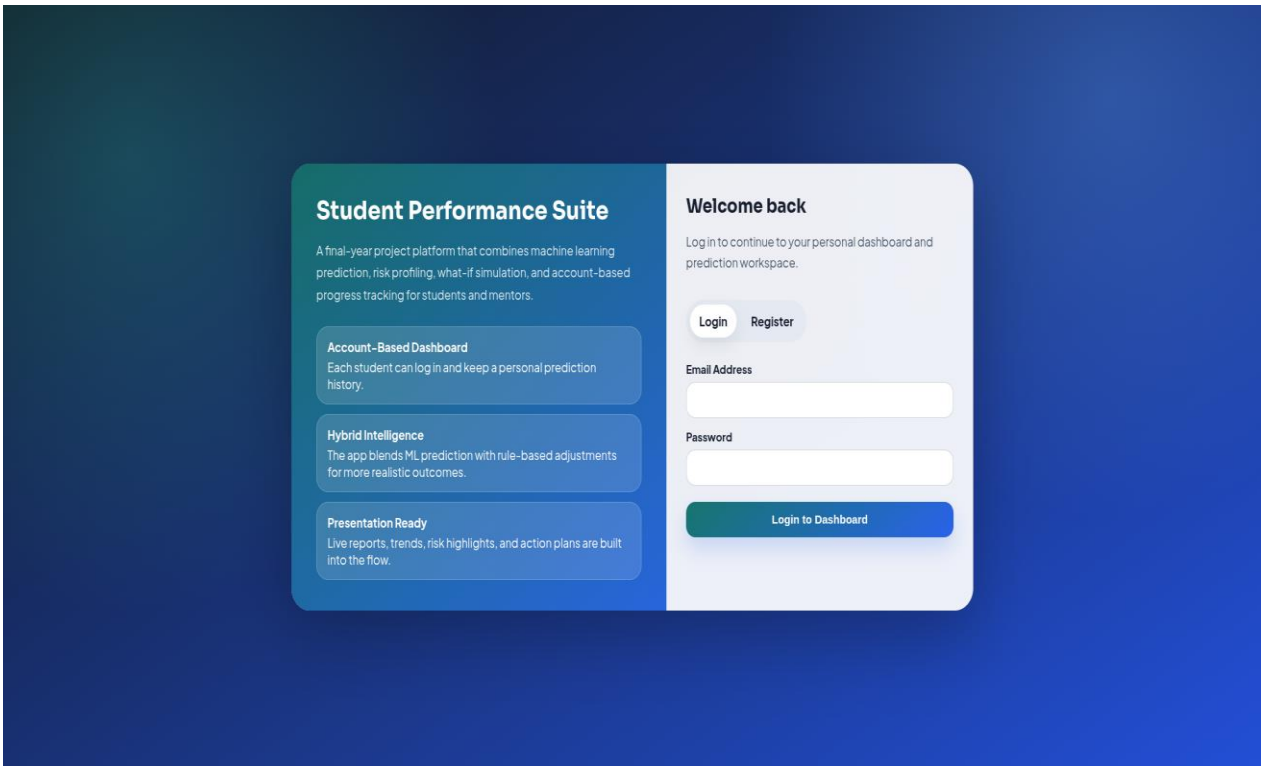
ACTIVITY DIAGRAM:-



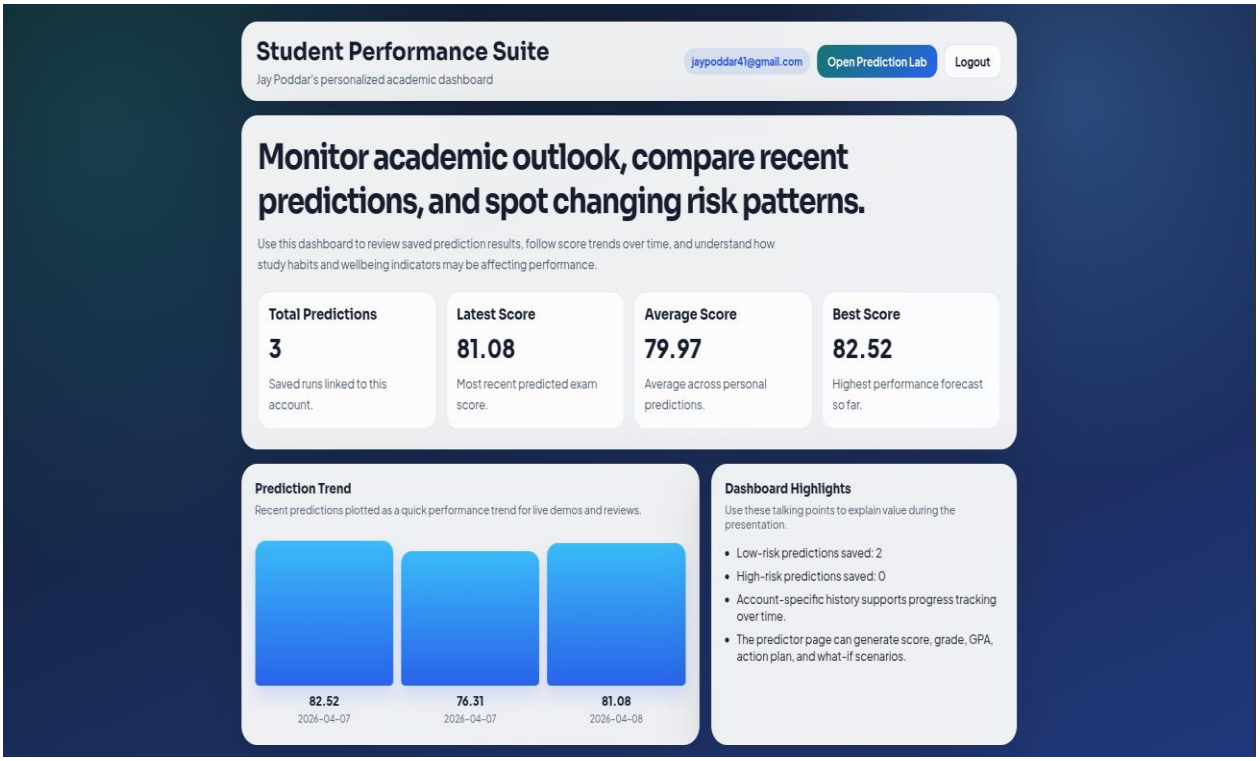

Adjustment Layer
Generate Feedback
Save History
Display Result
END

OUTPUT SCREENS

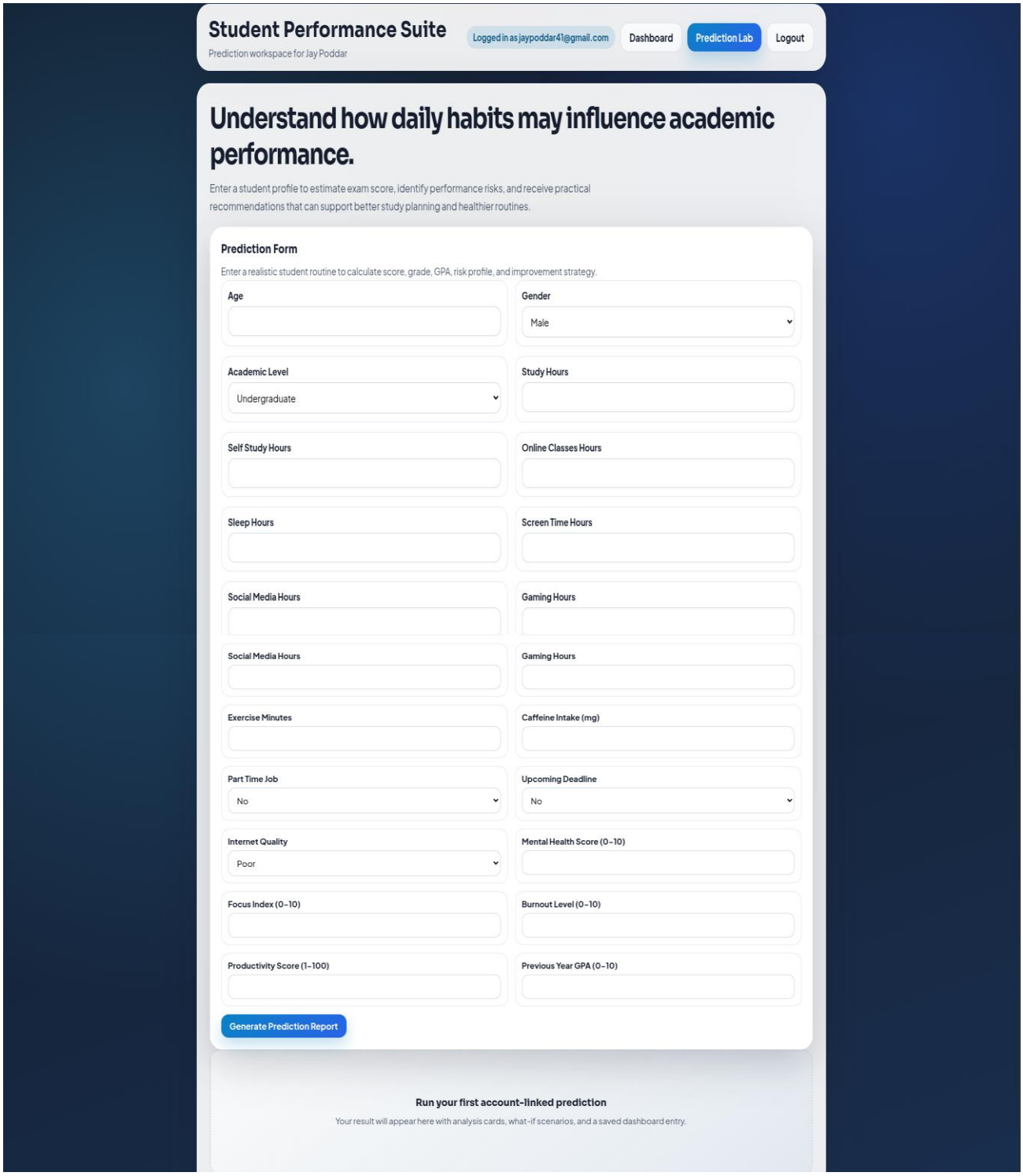
REGISTER PAGE



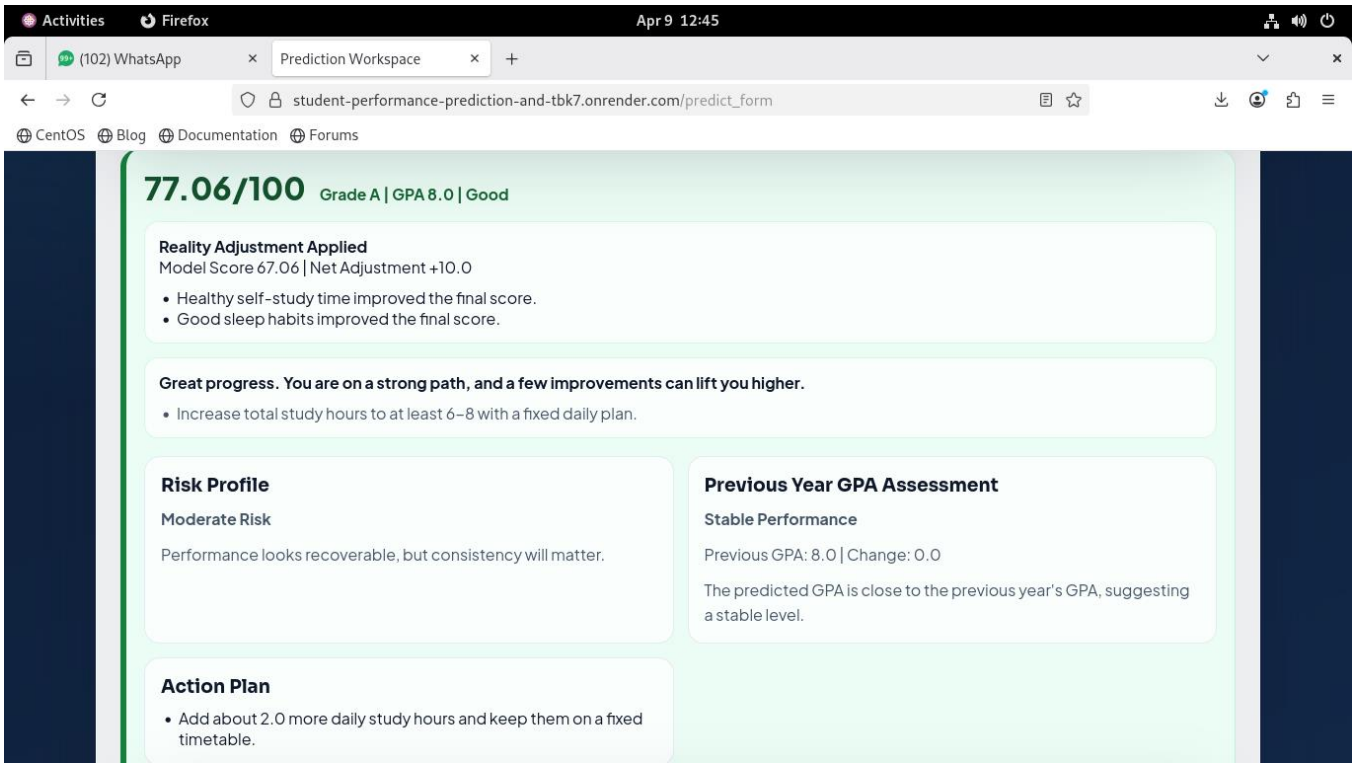
DASHBOARD

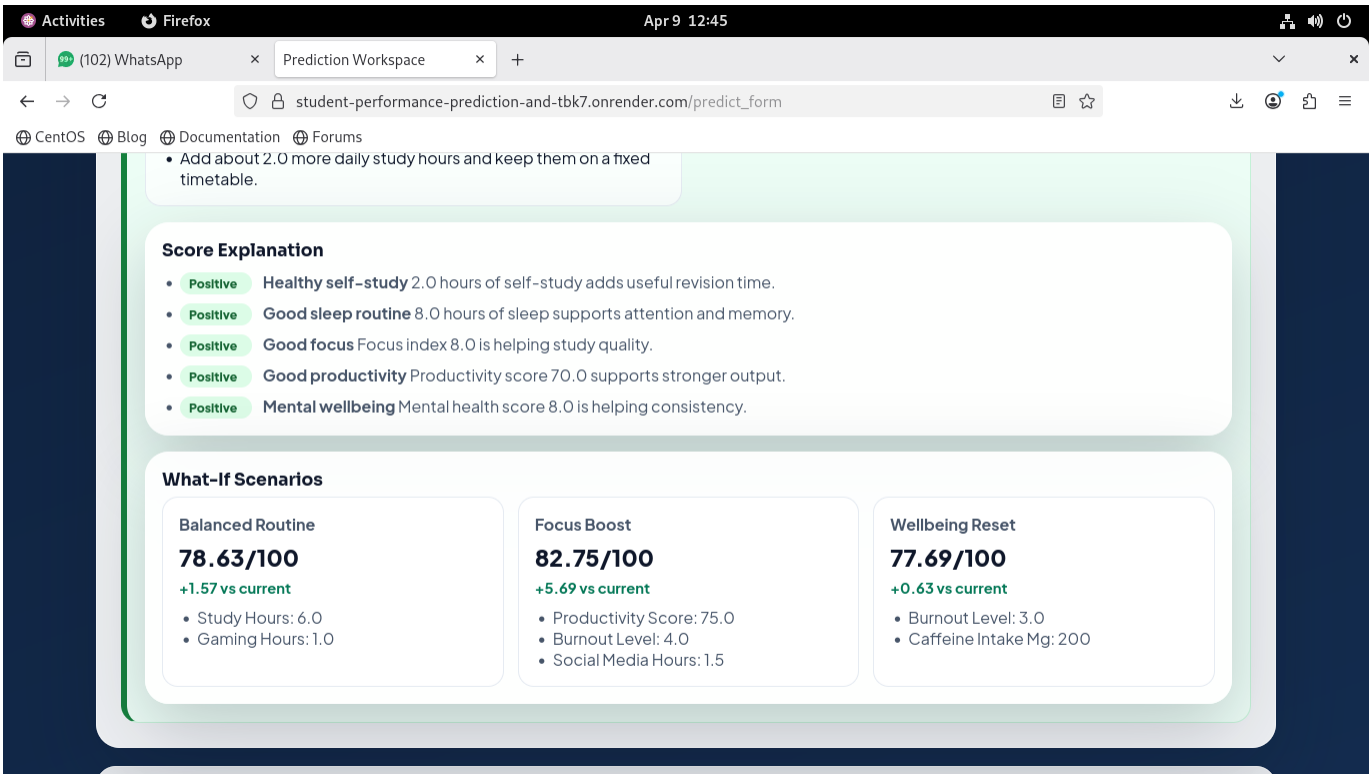


PREDICTION/OUTPUT PAGE

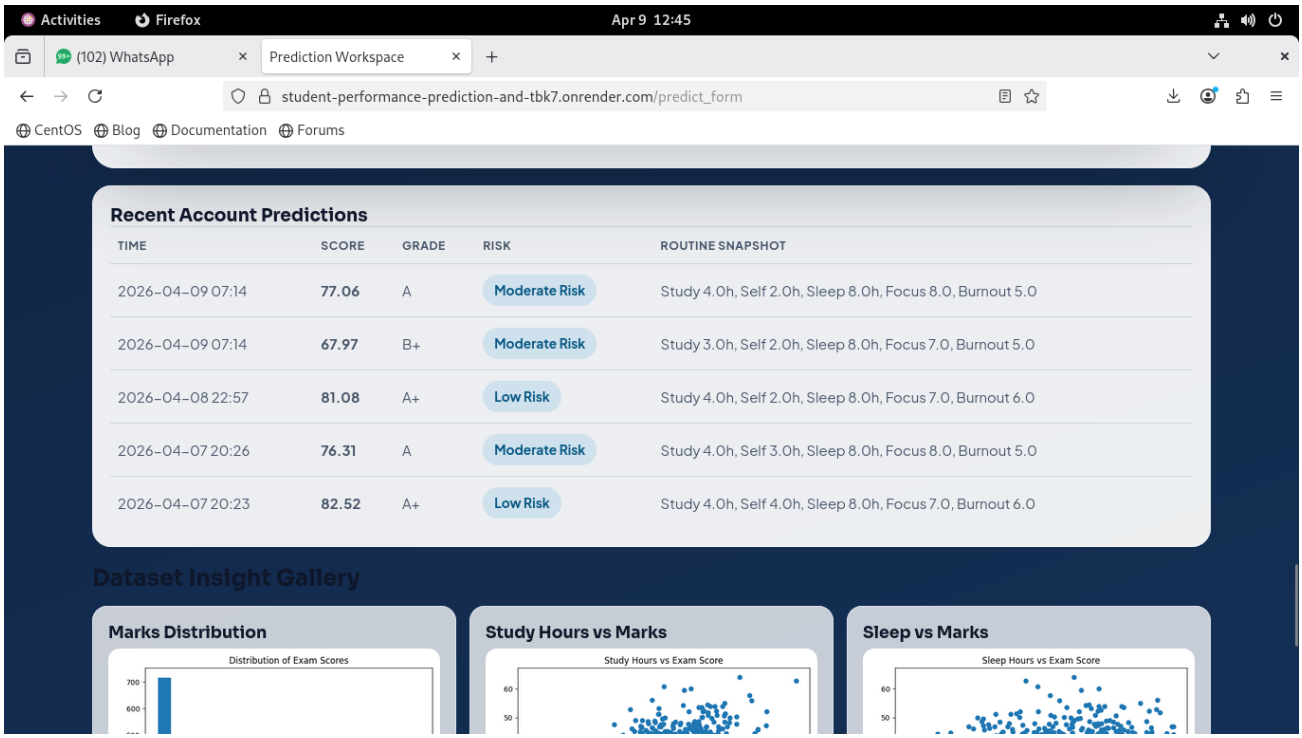


PREDICTION RESULT:-





REPORT & HISTORY:-



DATA INSIGHTS:-

FUTURE SCOPE

The Student Performance Prediction and Analysis System can be improved in several ways to make it even more powerful and useful for students and institutions:

1. **Advanced AI Integration** – Use deep learning models to improve prediction accuracy.
2. **Mobile Application** – Develop an Android/iOS app for easier access.
3. **Student Profiles & Login System** – Track performance over time for each student.
4. **Graphical Dashboards** – Include visual charts for performance trends.
5. **Explainable AI (XAI)** – Add explanation for predictions to make results understandable.
6. **Institution Integration** – Connect with college databases for automated data input.
7. **Personalized Study Plans** – Suggest tailored study schedules based on habits.
8. **What-if Analysis** – Allow students to simulate different study scenarios and see potential outcomes.
9. **Long-term Performance Analytics** – Monitor performance across semesters to identify improvement areas.
10. **Enhanced Feedback System** – Add motivational tips, reminders, and alerts for students at risk.

CONCLUSION

The Student Performance Prediction and Analysis System is a robust tool designed to help students understand and improve their academic performance.

Key takeaways:

- The system combines **machine learning with a rule-based adjustment layer** to provide realistic predictions.
- Students can input daily habits and receive predictions, grades, GPA, and personalized feedback.
- The system validates data, predicts exam scores, and stores history for long-term analysis.
- It acts as a **decision-support tool**, helping students improve study habits and performance.

Final statement:

> "This system not only predicts academic outcomes but also empowers students to make informed decisions, ensuring better performance and self-awareness."

BIBLIOGRAPHY

Books

1. Python Programming – McGraw Hill
2. Introduction to Machine Learning – McGraw Hill

Websites

1. Flask Official Documentation – https://flask.palletsprojects.com
2. Scikit-learn Documentation – https://scikit-learn.org
3. W3Schools – https://www.w3schools.com
4. Stack Overflow – https://stackoverflow.com

Research Papers / Articles

1. Breiman, L. (2001). Random Forests. *Machine Learning Journal*.
2. Kotsiantis, S. (2007). Supervised Machine Learning: A Review of Classification Techniques. *Informatica*.